MIL-P-50570A (AR)
19 July 1979
SUPERSEDING
MIL-P-50570 (MU)
14 July 1971

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, ROCKET ASSISTED M549 & M549A1, DELAY ASSEMBLY FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the delay assembly for the 155MM Rocket Assisted Projectile, M549 and M549A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-463 - Ethyl Alcohol, For Ordnance Use.
MIL-C-45662 - Calibration System Requirements.
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

MILITARY

MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9227721 - Box, Fiberboard, Packing, Ammunition, For Delay Assy, 155MM, M549
9227722 - Box, Packing, Ammunition, For Delay Assy, 155MM, M549
9235983 - Delay Assembly 155MM, M549

INSPECTION EQUIPMENT DRAWINGS

9225582 - Delay Testing Assembly
9225583 - Delay Housing Adapter
9225584 - Shell, Modified, Delay Test
9225585 - Operating Instructions
9246401 - Nitrogen Test Assembly
9246420 - Envelope Drawing (Porosity Test)
9278661 - Spotting Composition, Delay Test, M549

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9235983 and with all requirements specified in applicable specifications and standards.

3.3 Pyrotechnic composition. The flash, delay and igniter composition shall comply with applicable drawing requirements.

3.3.1 Particle size. The granulation of the tungsten delay composition, prior to loading or pelletizing shall be as specified in Table I when tested in accordance with Paragraph 4.4.3.6.3.

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TABLE I

USS SIEVE NO.	PERCENT RETAINED
20	3.0 Max.
80	91.0 Min.

3.3.2 Total volatiles. The total volatiles of the flash, delay and igniter composition at the time of loading shall not exceed 0.10 percent when tested in accordance with 4.4.3.2.

3.4 Flight test

3.4.1 Delay composition acceptance

3.4.1.1 Plus 70°F. After conditioning at 70°F for a period of 24 hours minimum, (min) the delay shall be flight tested at a proving ground as specified in 4.4.3.3.1.1. The average delay time shall be 7.75 seconds (sec) maximum (max) and 6.25 sec. min. The max. allowable standard deviation is 0.5 sec.

3.4.1.2 Plus 145°F. After conditioning at +145°F for a period of 24 hours min, the delay shall be flight tested at a proving ground as specified in 4.4.3.3.1.2. The average delay time shall be 5.25 sec. min., and 6.75 sec. max. The max. allowable standard deviation is 0.5 sec.

3.4.2 Delay assembly acceptance

3.4.2.1 Plus 145°F. After conditioning at 145°F for a period of 24 hours minimum, the delay assembly shall be flight tested at a proving ground (see 4.4.3.3.2.1). The average delay time shall be 6.00 seconds plus or minus .75 seconds and the standard deviation shall be no greater than .5 seconds.

3.4.2.2 Minus 65°F. After conditioning at minus 65°F for a period of 24 hours minimum, the delay assembly shall be flight tested at a proving ground (see 4.4.3.3.2.2). The average delay time shall be 8.50 seconds plus or minus .75 seconds and the standard deviation shall be no greater than .8 seconds.

3.4.2.3 Plus 70°F. After conditioning at 70°F for a period of 24 hours min, the delay assembly shall be flight tested at a proving ground (see 4.4.3.3.2.3). The average delay time shall be 7.90 sec. max. and 6.10 sec. min. The max. allowable standard deviation is 0.5 sec.

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3.5 Crimp strength. The washer crimp shall be capable of withstanding a 650 pound minimum load when tested as specified in 4.4.3.4.

3.6 Pyrotechnic porosity test. The pressure recorded at the exit end of the delay assembly, 80 seconds after application of the nitrogen gas, shall not be greater than 17 PSIG above the zero reference. The pressure recorded at the exit end of the delay assembly 110 seconds after application of the nitrogen gas shall be no greater than 27 PSIG above the zero reference. The test shall be conducted in accordance with 4.4.3.5.

3.7 Radiographic inspection. The delay column shall not be missing or be short in length and shall be free of voids. The igniter and/or flash composition shall not be missing. The boron potassium nitrate charge shall not be missing or contain voids greater than 1/16 square inch maximum when tested in accordance with 4.5.3.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements for workmanship are as shown on the applicable drawing and in the applicable specification and the following:

3.9.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications required thereon. Where cleaning and preparation are required by assembly details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.9.2 Foreign matter. Parts or assemblies shall not contain dirt, grease, chips, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Delay Assembly	9235983	30
Delay Housings/Crimp Test	9235983	5/Machine
Delay Assembly/For Ballistic Test	9235983	30

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic tests

4.3.2.1.1 Submission. Thirty delay assemblies packed in the format that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and Ballistic Testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1) and shall consist of thirty assemblies.

4.3.2.1.2 Testing. Proving Ground tests shall be performed on thirty (30) assemblies in accordance with 4.4.3.3.2, utilizing the procedures of 4.5.1.2 to determine compliance with 3.4.2.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Delay Assembly and Components	1 OF 1	9235983
			NEXT HIGHER ASSEMBLY: 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Assembly				
	Examination for Defects	30	-	3.2	4.4.2.3, 4.4.2.4, 4.4.2.5
	Porosity Test	30	-	3.6	4.4.3.5
	Radiographic Test for Proper Loading	30	-	3.7	4.5.3
	Delay Housing/Crimp Test				
	Crimp Strength	5/Machine	-	3.5	4.4.3.4

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded delay assemblies shall contain:

a. Inert parts from one interfix lot number from one manufacturer.
b. Flash composition from not more than one inspection lot.
c. Delay composition from not more than one inspection lot.
d. Igniter composition from not more than one inspection lot.
e. Boron Potassium Nitrate pellets from not more than one interfix lot.

4.4.1.1 Delay composition lot. A delay composition lot is defined as that quantity of delay material from which representative samples have been selected, loaded, and forwarded to a proving ground for acceptance. A delay composition lot may be formed by any of the following methods:

a. Blending two or more batches of delay powder or pellets together to form a delay composition lot.
b. Designating a batch of delay powder or delay pellets as delay composition lot.

Once samples are selected from the batch or delay composition lot, the batch or lot shall not be subsequently blended.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Delay Assembly, Prior to Loading Delay Composition	1 OF 1	9235983
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Flash composition on sidewall		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.2	Delay Assembly, Prior to Loading Igniter Composition		1 OF 1		9235983
					NEXT HIGHER ASSEMBLY 9235999
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance end of housing to delay composition		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50570A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Delay Assembly, After Crimping Washer		9235983
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Washer missing		100%	3.2	Visual
2	Crimp missing or incomplete		100%	3.2	Visual
3	Flash or igniter composition missing		100%	3.2	Visual
Major					
101	Depth of washer, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50570A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Delay Assembly, Prior to Assembly of Igniter Tube Plug		9235983
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Igniter charge missing		100%	3.2	Visual
Major					
101	Weight of igniter charge		0.40%	3.2	Balance
Minor					
201	Distance tube end surface to Boron Potassium Nitrate Pellets, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Delay Assembly		9235983
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Igniter tube plug missing or improperly positioned		100%	3.2	Visual
2	Crimp on tube missing or body cracked by crimping operation		100%	3.2	Visual
Major					
101	Outside diameter of tube, max.		100%	3.2	Gage
Minor					
201	Identification missing or improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.9	Visual

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4.4.3 Testing

4.4.3.1 Consolidation load - Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. For verification each consolidation machine shall be calibrated for consolidation load and dwell time at the beginning of each production shift or portion thereof using a load cell and stopwatch. In addition, a suitable warning device shall be assembled to each consolidation machine and calibrated at the same time as the consolidation machine. If, during the loading of the delay the pressure drops before the applicable consolidation load requirement or the dwell time is short of the drawing requirement, that delay shall be rejected.

4.4.3.2 Total volatiles - Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified total volatiles requirement. For verification a sample of delay, igniter and flash composition from each 8 hours production and from each new container shall be selected to perform a total volatiles analysis. If any sample fails to meet the requirements of 3.3.2, the 8 hours production represented by the sample or the new container of material shall be rejected. The sample shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.4.3.3 Flight test

4.4.3.3.1 Pyrotechnic material acceptance

4.4.3.3.1.1 Plus 70°F - Major defect. The acceptance criteria shall be in accordance with MIL-STD-414, Section B, Table B3, code letter F (10 samples), and an AQL of 6.5%. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of paragraph 3.4.1.1 is not met - Major defect, or if the delay assembly lot fails to ignite - Critical defect, or ignites in the tube - Critical defect, the delay assembly lot shall be rejected. The test shall be conducted in accordance with paragraph 4.5.1.1.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance.

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4.4.3.3.1.2 Plus 145°F. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of paragraph 3.4.1.2 - Major defect, or if any delay functions in the tube - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.1.2. A minimum of 8 radar delay times shall be required to determine delay time acceptance.

4.4.3.3.2 Delay assembly lot acceptance

4.4.3.3.2.1 Plus 145°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.4.2.1 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.2.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance.

4.4.3.3.2.2 Minus 65°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.4.2.2 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.2.2. A minimum of 8 radar delays times shall be required to determine delay time acceptance.

4.4.3.3.2.3 Plus 70°F. The acceptance criteria shall be in accordance with MIL-STD-414, Section B, Table B3, code letter F (10 samples), and an AQL of 10%. The sample delay assemblies shall be selected at random and forwarded to a Government proving ground. If the sample fails to meet the requirements of 3.4.2.3 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.2.3. A minimum of 8 radar delay times shall be required to determine delay time acceptance.

4.4.3.4 Crimp strength - Major defect. At the beginning of each production shift or portion thereof, an inert delay housing shall be crimped on each crimping device using the same technique as in regular production. After crimping the delay housing shall be

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sectioned perpendicular to the longitudinal axis and the load applied axially to the crimp from the sectioned end. A tool manufactured to the same diameter as the delay washer shall be used in applying load to the crimp. If the crimp fails to meet the applicable load requirements, the crimping device shall be adjusted until the crimp meets the requirements of 3.5. The process shall then be monitored to assure that proper crimp strength is maintained throughout production.

4.4.3.5 Porosity test - Major defect. This test shall be performed 100 percent until 1500 consecutive units have passed without failure. After the above criteria has been met, sampling shall be performed randomly at the rate of 18 delay assemblies for each shift production or portion thereof. If any one of the 18 delay assemblies fails to meet the requirements of paragraph 3.6, the total shift production shall undergo the porosity test and an additional 1500 units tested to qualify sampling again.

4.4.3.6 Determination of average particle size

4.4.3.6.1 Molybdenum trioxide - Major defect. A sample shall be selected from each lot of molybdenum trioxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 9212470 for particle size, the lot shall be rejected.

4.4.3.6.2 Chromium oxide - Major defect. A sample shall be selected from each lot of chromium oxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 9212469 for particle size, the lot shall be rejected.

4.4.3.6.3 Tungsten delay composition - Major defect. The delay composition particle size shall be determined in accordance with Method 201.1 of MIL-STD-1234 except that a minimum 100 gram sample shall be used and the shaker operated, without striker, for one minute.

4.4.3.7 Radiographic test for proper loading - Major defect. Loaded delay assemblies shall be inspected in accordance with 4.5.3 on a 100 percent basis. Any assembly not meeting the requirements of 3.7 shall be rejected and removed from the lot.

4.4.3.8 Vinyl Alcohol Acetate Resin (VAAR) solids determination. The concentration of solids in the VAAR solution may change during storage due to evaporation of solvent. VAAR solids content in the delay blend can only be obtained if the solution to

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solids weight ratio is known. Prior to blending the delay composition, a sample of VAAR solution shall be tested to determine this ratio. If the sample fails to meet the drawing requirements, the batch of VAAR shall be rejected. The sample shall be tested in accordance with MIL-V-50433.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract, see Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Flight test

4.5.1.1 Pyrotechnic material acceptance

4.5.1.1.1 Plus 70°F. The loaded delay assemblies shall be radiographed after pressing into the delay adapter and assembled to a 155mm modified M101 shell as specified in Drawing 9225582. The Spotting Charge shall consist of 15 grams of Spotting Charge FW176 in accordance with Drawing 9278661.

A dummy fuze and rocket off cap shall be assembled to the test vehicle prior to temperature conditioning. The complete round, including propelling charge shall be conditioned at 70°F plus or minus 5°F for 24 hours minimum. The complete round shall be removed from the conditioning chamber, the rocket selector cap removed, and the round fired from an M199 Cannon within 3 minutes after removal from the conditioning box. A Zone 8 M203 Series propelling charge shall be utilized in this test. A smear camera shall be mounted near the muzzle to photograph the projectile in order to determine the presence of the delay as the projectile exits the muzzle. Delay time shall be determined by use of radar or an IR Sensor Tracor (Jefferson PG manufacture Detector Model No. ROA2Y or equivalent). Stopwatch times may be used as a back-up to confirm that the delay functioned but shall not be used in computations for delay acceptance. If the delay expulsion was not observed by radar, infrared or visually and it has been determined that the delay was present at the muzzle, the shell shall be recovered to determine if the delay functioned.

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4.5.1.1.2 Plus 145°F. This test shall be conducted as specified in 4.5.1.1.1 except that the conditioning temperature for the delay assembly and projectile shall be plus 145°F and a proof charge conditioned at plus 70°F is to be used in lieu of a M203 Series propelling charge.

4.5.1.2 Delay assembly lot acceptance

4.5.1.2.1 Plus 145°F. This test shall be conducted as specified in 4.5.1.1.1 except that the conditioning temperature for the delay assembly and projectile shall be plus 145°F and a proof charge conditioned at plus 70°F is to be used in lieu of a M203 Series propelling charge.

4.5.1.2.2 Minus 65°F. This test shall be conducted as specified in 4.5.1.1.1 except that the conditioning temperature for the delay assembly and projectile including M203 propelling charge, shall be minus 65°F.

4.5.1.2.3 Plus 70°F. This test shall be conducted as specified in 4.5.1.1.1.

4.5.2 Porosity test (see 6.5). Each delay assembly shall be conditioned at plus 145°F and assembled to the porosity test equipment. Nitrogen gas under a pressure of 1000 PSIG plus or minus 50 PSIG shall be applied to the delay column at the threaded end of the housing. The pressure at the exit end of the delay column shall be recorded at 0.5 seconds, 80 seconds and 110 seconds. The pressure reading at 0.5 seconds shall be recorded and used as the zero reference.

4.5.3 Radiographic test for proper loading. The sample delays shall be subjected to radiographic examination and shall be oriented so as the two distinct delay columns do not overlap on the x-ray. In all views the radiographic beam shall be perpendicular to the longitudinal axis of the delay assembly. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 6.7). In addition, the sensitivity of the x-ray technique shall be 2 percent and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.4.

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4.5.4 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive train. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. (Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units). Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Drawings 9227721 and 9227722.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 155MM Rocket Assisted Projectile, M549 and M549A1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, N.J. 07801.

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6.4 Submission of test data. In addition to the normal distribution of records, one copy of all ballistic data shall be forwarded to:

Commander
ARRADCOM
ATTN: DRDAR-QAR-Q/DRDAR-LCU-E-F
Dover, New Jersey 07801

6.5 Handling of delay assemblies. Precaution should be taken during loading, intraplant shipment for in process test, subsequent to loading and prior to packing to assure delays are not exposed to moisture. Delay assemblies should not be removed from moisture-proof packing until such time as the units are to be placed in the conditioning chamber for porosity testing and flight testing.

6.6 Pilot mixture - Advisory. The burning time of delay compositions manufactured from different lots or supplies of tungsten may vary. Pilot mixtures are therefore advisory to define the composition burning time curve for new lots of supplies of the tungsten.

6.7 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q for review and approval upon implementation at each 6 month period of production.

6.8 Delay composition - For information. To insure that the drawing requirements are met, a chemical analysis should be performed to determine the percent of tungsten and barium chromate in the delay composition before loading.

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6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No.: 1320-A 494

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